Vivekanand Education Society's Institute of Technology

Department of Computer Engineering

Group No.: 11

Date :- 16 / 09 / 2025

Project Synopsis(2025-26) - Sem VII

AutoGenie: Agentic AI Agent for Video Pre-Production Planning & Asset Sourcing

Dr. Sharmila Sengupta
Professor, CMPN

Jennifer Thozhuvathinkal	**Anusha Gonal**	**Varun Budhani**	**Shantanu Bhosale**
V.E.S.I.T	V.E.S.I.T	V.E.S.I.T	V.E.S.I.T
2022.jennifer.thozhuvathinkal@ves.ac.in	2022.anusha.gonal@ves.ac.in	2022.varun.budhani@ves.ac.in	2022.shantanu.bhosale@ves.ac.in

Abstract

This project introduces an AI Agent for Video Pre-Production Planning and Asset Sourcing, designed to streamline the creative workflow for content teams. Traditional pre-production processes are time-intensive, involving manual transformation of briefs into structured plans, asset gathering, and ensuring license compliance, often leading to inefficiencies and creative bottlenecks. The proposed agentic AI system addresses these challenges by leveraging autonomous, collaborative agents that unify prompt interpretation, video classification, planning, and asset management into a cohesive, iterative platform.

At its core, the system parses user briefs to classify video types, generate shot/asset plans with timing constraints, and decide per-asset whether to source from the web (via search, crawling, filtering, and licensing checks) or generate anew (using model APIs for images, videos, and audio). A conversational chat interface enables rapid refinements with versioned diffs, while modular tools ensure coherence, precision, and legal safety. Built on a React frontend, LLM-orchestrated backend, and data layers like Postgres and vector stores, the platform outputs production kits including storyboards, asset inventories, and exports.

By automating and democratizing pre-production, this agentic solution enhances efficiency, creativity, and accessibility for creators, reducing turnaround times and fostering scalable video content development without requiring deep technical expertise.

Introduction

Artificial Intelligence (AI) has become a transformative force across industries, enabling innovative applications in education, healthcare, finance, entertainment, and more. However, developing AI-driven systems remains a complex, resource-intensive process, requiring expertise in model integration, data handling, multimodal processing, and deployment. These challenges create significant barriers for individuals, startups, and organizations seeking to leverage AI for creative and practical solutions.

To address this gap, we propose an Agentic AI Platform tailored for video pre-production planning and asset sourcing, designed to streamline content creation workflows. The platform empowers users to transform creative briefs into structured, license-safe video plans with minimal technical expertise. By leveraging autonomous, collaborative AI agents, it unifies prompt interpretation, video type classification, shot/asset planning, and automated sourcing or generation of assets (images, videos, audio) via web crawling, license filtering, or model APIs. A conversational interface facilitates iterative refinements with versioned diffs, ensuring transparency and efficiency in plan evolution. The system outputs comprehensive production kits, including storyboards, asset inventories, and exportable formats like PDF and JSON.

Built on a modular architecture with a React frontend, LLM-orchestrated backend, and robust data layers (Postgres, vector stores), the platform ensures scalability,consistency, and legal compliance. By abstracting complexities into intuitive, reusable components, this solution democratizes AI-driven video pre-production, enabling creators to produce high-quality content efficiently and fostering innovation across diverse use cases.

Problem Statement

Content creation teams face significant challenges in the pre-production phase of video development, where transforming a creative brief into a structured, license-compliant plan demands excessive time and resources. The process involves multiple complex tasks: classifying the video format, defining shot sequences and asset requirements with precise timing, deciding whether to source assets from external repositories or generate them anew, and ensuring all assets align with brand identity and legal licensing requirements. These tasks are often manual, fragmented, and prone to inefficiencies, leading to delays and increased costs. Additionally, iterating on plans requires cumbersome communication, and sourcing or generating assets lacks streamlined automation, further complicating the workflow.

To address these challenges, there is a need for an AI-driven agent that unifies and automates the pre-production process. This agent should interpret user prompts to classify video types and enforce format constraints, generate detailed shot and asset plans, make intelligent source-versus-generate decisions for each asset, and provide automated tools for web-based asset sourcing (with search, crawling, deduplication, and licensing filters) and safe asset generation (via model APIs for images, videos, and audio). A conversational interface is essential for iterative plan refinement with clear versioned diffs, enabling rapid convergence on a final plan. The solution must deliver a comprehensive production kit, including a storyboard PDF, CSV/JSON asset lists, prompts, and links, while ensuring quality through coverage, coherence, retrieval precision, legal compliance, and user satisfaction. By streamlining these processes, the agent aims to reduce pre-production time, enhance creative efficiency, and make video content creation accessible to users with minimal technical expertise.

Proposed Solution

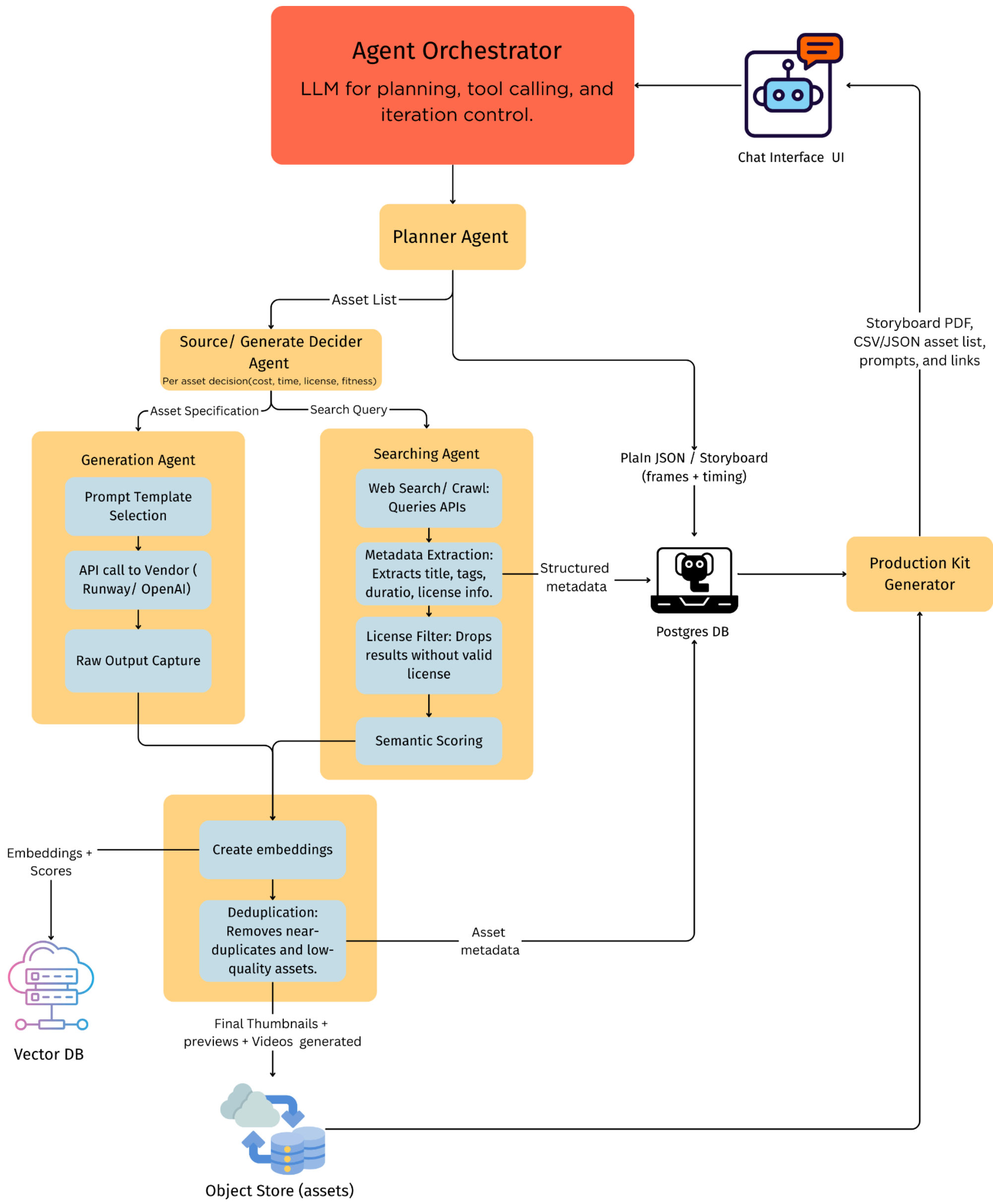
The proposed system is an Agentic AI Platform tailored for video pre-production planning and asset sourcing. Instead of handling tasks manually, the platform employs collaborative AI agents to automate interpretation, planning, sourcing, and compliance.The system will comprise the following modules:

- **Brief Interpretation Module** – Parses creative briefs to extract requirements such as platform, aspect ratio, duration, and style.
- **Video Classification Module** – Identifies the target video type and applies format-specific constraints.

- **Shot & Asset Planning Module** – Generates scene-level shot lists with timing, transitions, and required assets.
- **Asset Sourcing & Generation Module** – Decides whether to source assets from the web (with crawling, deduplication, and license filtering) or generate them using model APIs (images, video, audio, TTS).
- **Conversational Interface Module** – Provides a chat-based React frontend for iterative refinements, version tracking, and user-friendly collaboration.
- **Data & Storage Module** – Uses PostgreSQL for structured plans and a vector store for semantic asset retrieval.
- **Export Module** – Produces comprehensive production kits including storyboards (PDF), asset inventories (CSV/JSON), and downloadable resources.

By combining these modules, the platform streamlines pre-production into a cohesive workflow, ensuring efficiency, creativity, and legal compliance while lowering the technical barrier for creators.

Methodology / Block Diagram



Hardware , Software and tools Requirements

- **Programming Languages:** Python (for the backend), JavaScript (for the React frontend).
- **AI Frameworks: LangChain**, **AutoGen**, or **CrewAI** for orchestrating the multi-agent system.
- **Databases: PostgreSQL** for storing plans and assets, and a **Vector Store** (like Qdrant or Milvus) for semantic retrieval.
- **Web Frameworks: React** (frontend) and a backend orchestrated by an LLM.
- **Cloud Services:** Cloud infrastructure for hosting a web app, orchestrator, and data layers.
- **APIs: LLM APIs** for planning and tool calling, and vendor APIs for image/video/TTS (Text-to-Speech) generation.
- **Web Tools:** Web search and crawling tools for sourcing assets from the web.
- **Tools: FFmpeg** for video processing (implied by "video editing" being out of scope, but processing for exports is a possibility). The system also requires a conversational chat interface for rapid refinements and an export function for production kits.

Proposed Evaluation Measures

- **Task Success Rate:** This measures the percentage of user briefs that the AI agent successfully transforms into complete and usable video pre-production kits, including shot lists, storyboards, and asset inventories. A higher success rate indicates the system's ability to accurately interpret and execute complex user requests.
- **Time Savings & Efficiency:** This metric quantifies the reduction in time required for pre-production planning and asset sourcing compared to traditional manual methods. It would be measured by comparing the time taken to complete a task with and without the agentic system.
- **Asset Licensing Compliance:** This evaluates the system's effectiveness in sourcing assets with proper licensing. The measure would be the percentage of sourced assets that pass a compliance audit, ensuring legal safety and reducing copyright risks for creators.
- **User Satisfaction:** This gauges the overall user experience through surveys or feedback forms, focusing on the conversational interface, the quality of the generated plans, and the ease of refinement. Higher satisfaction scores indicate the system is intuitive and valuable to creators.

Conclusion

This project aims to simplify and accelerate the often time-consuming pre-production phase of video creation. By combining the strengths of AI-driven planning, automated sourcing, and safe content generation, the system acts as a supportive co-pilot for content creators and marketing teams. The solution not only improves efficiency but also ensures legal compliance, quality assurance, and creative flexibility. Through a conversational interface, users can iteratively refine their video plans, making the process more interactive and less overwhelming. In the long run, such a human-AI integrated system has the potential to become a standard tool in digital content workflows, bridging the gap between raw ideas and production-ready resources.

References

[1] Huang, Kaiyi, Yukun Huang, Xintao Wang, Zinan Lin, Xuefei Ning, Pengfei Wan, Di Zhang, Yu Wang, and Xihui Liu. "FilMaster: Bridging Cinematic Principles and Generative AI for Automated Film Generation." arXiv preprint arXiv:2506.18899 (2025).

[2] Venkatesh, Kavana, Connor Dunlop, and Pinar Yanardag. "Crea: A collaborative multi-agent framework for creative content generation with diffusion models." arXiv preprint arXiv:2504.05306 (2025).

[3] Lin, Han, Abhay Zala, Jaemin Cho, and Mohit Bansal. "Videodirectorgpt: Consistent multi-scene video generation via llm-guided planning." arXiv preprint arXiv:2309.15091 (2023).

[4] Zhang, Ruihan, Borou Yu, Jiajian Min, Yetong Xin, Zheng Wei, Juncheng Nemo Shi, Mingzhen Huang et al. "Generative AI for Film Creation: A Survey of Recent Advances." In Proceedings of the Computer Vision and Pattern Recognition Conference, pp. 6267-6279. 2025.

[5] Orak, Cihan, and Zeynep Turan. "Using artificial intelligence in digital video production: A systematic review study." Journal of Educational Technology and Online Learning 7, no. 3 (2024): 286-307.

[6] Liu, Zhuoer. "Analysis of the impact of artificial intelligence on the media and film industries." *Lecture Notes in Education Psychology and Public Media* 35 (2024): 219-223

[7] Izani, M., M. Fauzan, A. Razak, and Z. Kamarolzaman. "Lights, Camera, AI Action: Navigating the Disruptive Potential of Artificial Intelligence in Filmmaking."

[8] Jayanthiladevi, A., Arun Gnana Raj, R. Narmadha, Sajin Chandran, Sai Shaju, and K. Krishna Prasad. "AI in video analysis, production and streaming delivery." In *Journal of Physics: Conference Series*, vol. 1712, no. 1, p. 012014. IOP Publishing, 2020.

[9] Krishnan, Naveen. "Ai agents: Evolution, architecture, and real-world applications." *arXiv preprint arXiv:2503.12687* (2025).

[10] Zhou, Jiayi, Renzhong Li, Junxiu Tang, Tan Tang, Haotian Li, Weiwei Cui, and Yingcai Wu. "Understanding nonlinear collaboration between human and AI agents: A co-design framework for creative design." In *Proceedings of the 2024 CHI conference on human factors in computing systems*, pp. 1-16. 2024.